

What we're doing

Can Helico's own confidence rank native and near-native protein structures above Rosetta decoys when each candidate is converted to Helico's contact-map representation, and how does it compare with AF2Rank, DeepAccNet, and the Rosetta energy function on the exact AF2Rank benchmark?

Why

A candidate-derived contact map that is geometrically compatible with the target sequence should let contact-conditioned Helico produce a confident structure, whereas inconsistent decoy contacts should reduce Helico pTM. We therefore expect target-wise Helico pTM to correlate positively with candidate TM-score and recover high-quality candidates substantially better than Rosetta energy, but probably below AF2Rank because the contact map discards most of the template geometry available to AlphaFold. The native-versus-decoy discrimination question and the standard decoy-quality ranking question will both be reported.

Exact benchmark and execution

The exact public AF2Rank benchmark joins cleanly: 133 targets, 180,079 decoys, 133 natives, and corrected baseline outputs with no missing key. All 180,212 candidates ran on 96 independent CoreWeave H100 jobs; all jobs succeeded with no production retry. Wall time was 3 hours 35 minutes, with 314.45 inference H100-hours and 320.92 accounted H100-hours.

All candidates are retained. A systematic extra N-terminal lysine in 1,000 `1iib` decoys is uniquely mapped and explicitly removed rather than silently shifting indices or excluding candidates.

Complete decoy-ranking result

Pure Helico pTM reaches 0.841 mean target-wise Spearman correlation with candidate TM-score, above Rosetta at 0.759 but below AF2Rank at 0.925. Its mean top-1 TM-score is 0.879, below Rosetta's 0.901.

The secondary Helico composite (pTM times candidate/output TM-score) reaches 0.873 correlation and 0.936 mean top-1 TM-score. The latter is slightly above AF2Rank's 0.933 point estimate, but the paired 95% interval on the difference $[-0.0035, 0.0093]$ includes zero.

Complete native-versus-decoy result

Pure Helico pTM ranks the exact native first for only 4/133 targets and gives a mean native rank of 130.3. Confidence alone is therefore not an effective exact- native selector despite tracking overall decoy quality.

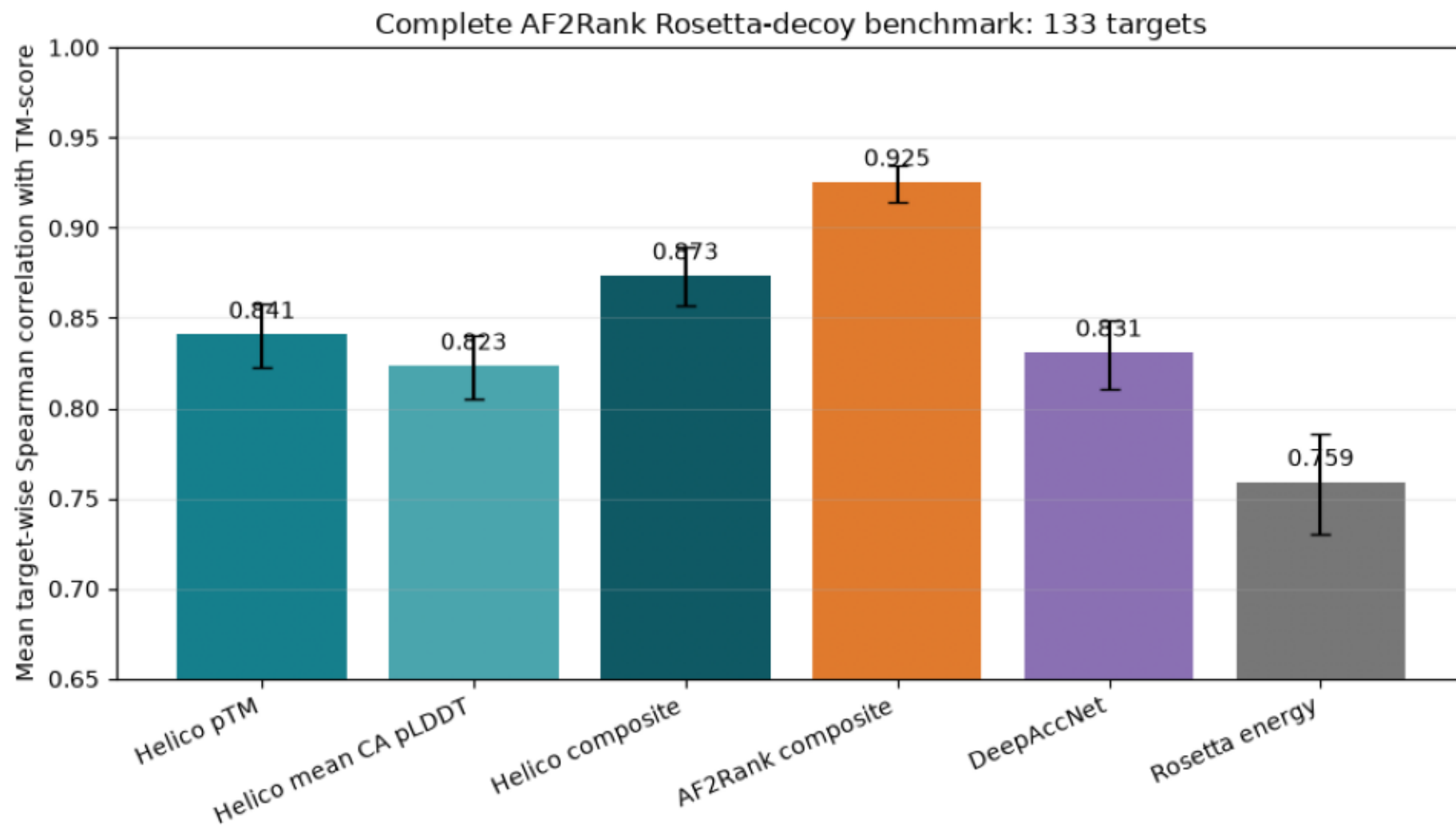
The Helico composite selects 51/133 natives and gives mean rank 26.5. AF2Rank selects 52/133 and gives mean rank 31.1. Paired target-bootstrap intervals for top-1 recovery, mean rank, and AUROC all include zero difference. DeepAccNet and Rosetta cannot be evaluated for native recovery because their native rows use `-1` sentinel scores.

Bottom line

The confidence-only hypothesis is only partially supported: Helico pTM ranks quality well but does not reliably identify the true native. Combining pTM with candidate/output structural agreement is competitive with AF2Rank for selecting the best candidate and the exact native, though it remains weaker for ranking the full decoy ensemble. Follow-up should validate the composite on independent decoy sets.

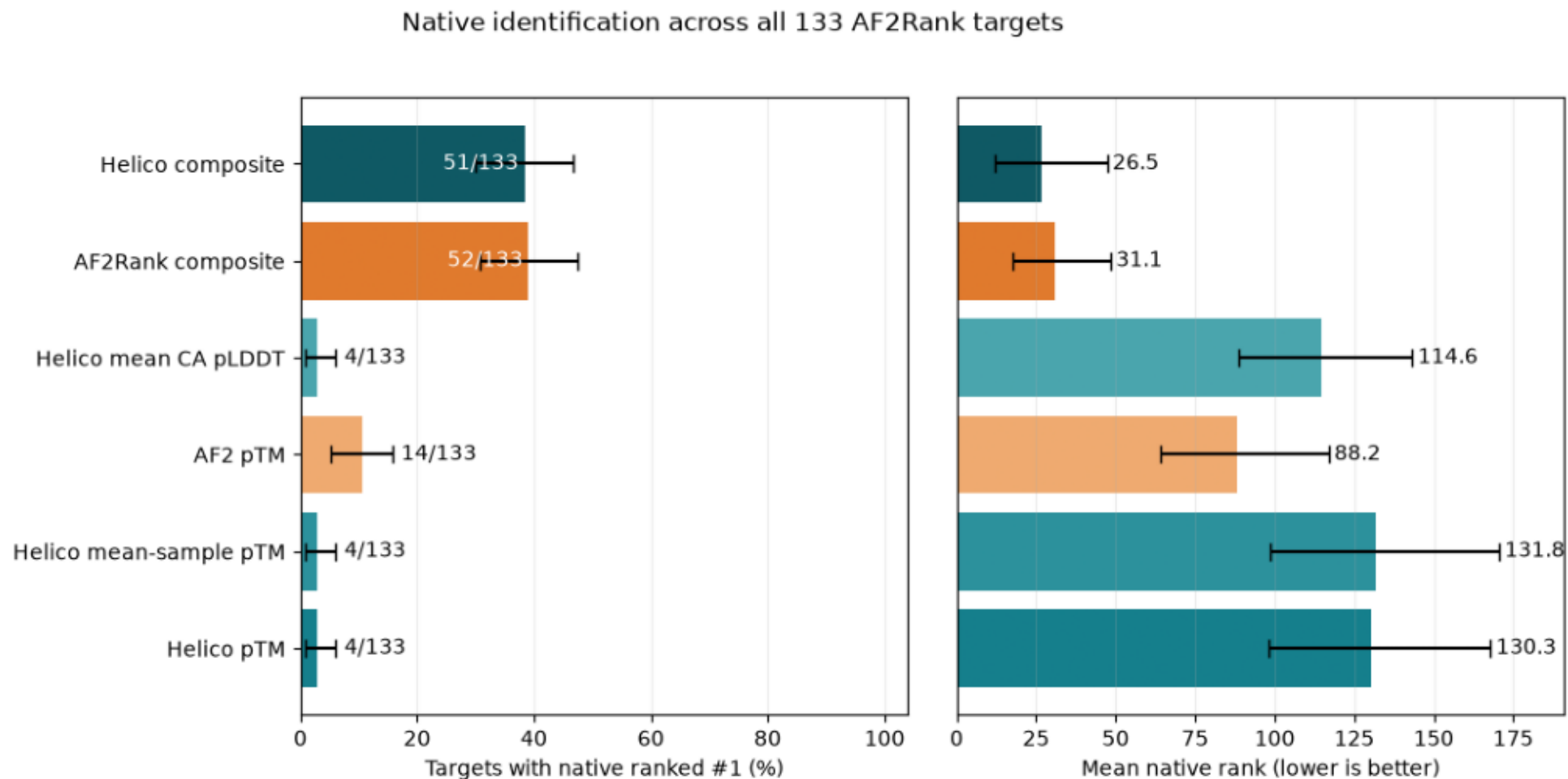
full_metric_comparison.png

Mean target-wise Spearman correlation across all 133 AF2Rank targets; error bars are 95% target-bootstrap intervals.



full_native_selection.png

Native top-1 recovery and mean native rank over the complete benchmark; error bars are 95% target-bootstrap intervals.

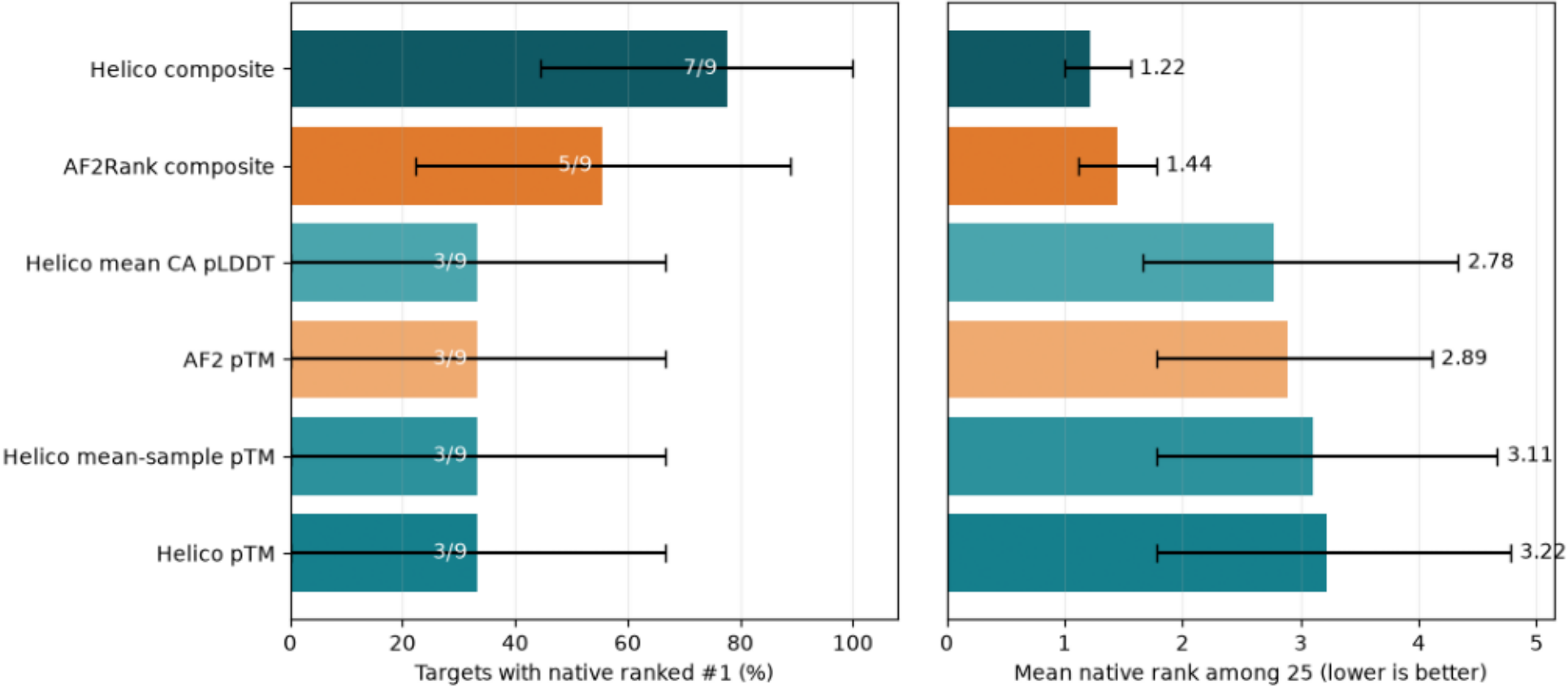


Error bars: 95% target-bootstrap interval. DeepAccNet and Rosetta omitted: their AF2Rank native rows contain -1 sentinel scores.

pilot_native_selection.png

Native top-1 recovery and mean native rank across the nine-target pilot. Each target contains its native and 24 TM-stratified decoys. DeepAccNet and Rosetta cannot be scored for native recovery because their corrected AF2Rank native rows contain -1 sentinels.

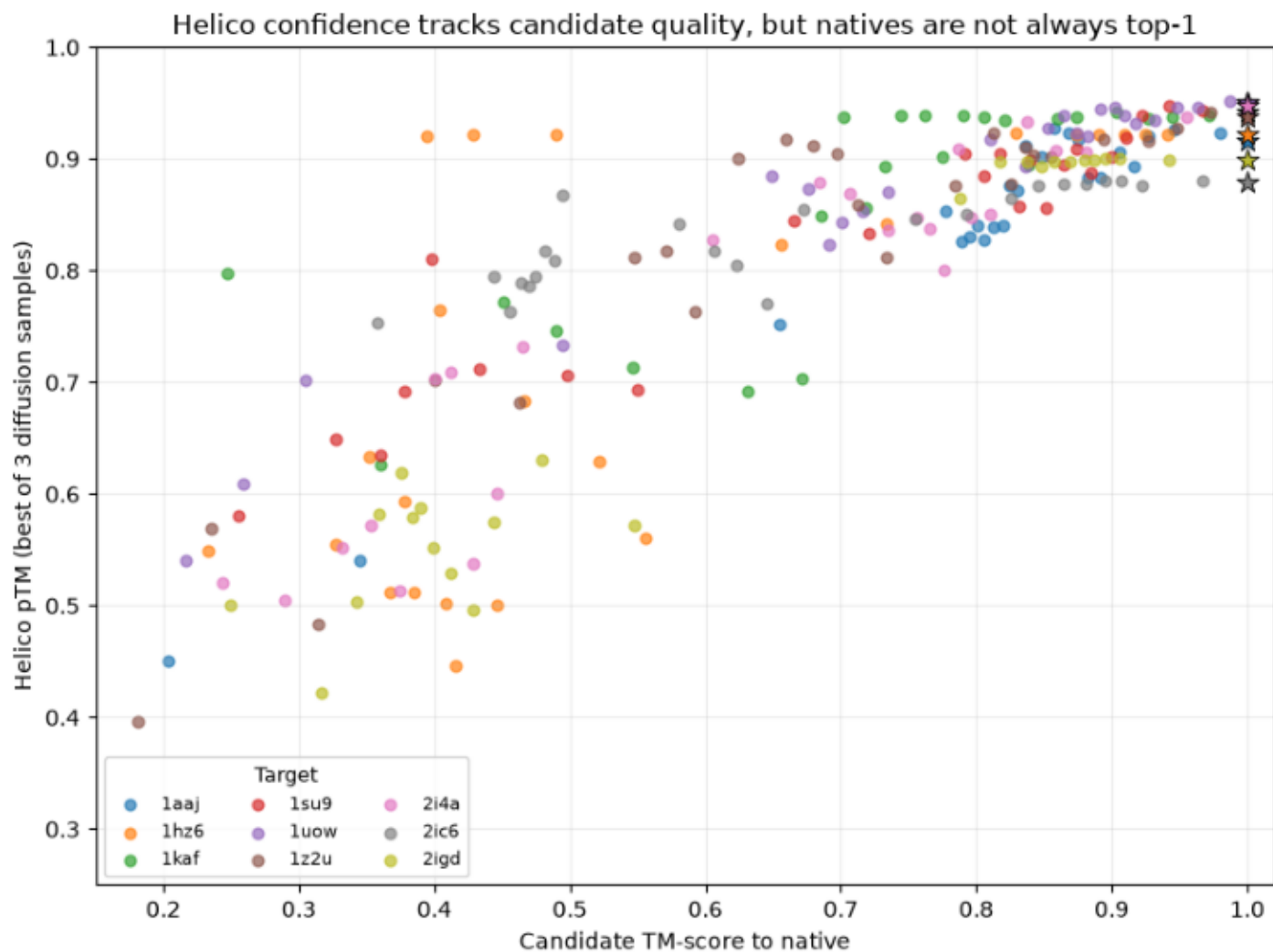
Native identification in the pilot: 9 targets, 24 decoys + native each



Error bars: 95% target-bootstrap interval. DeepAccNet and Rosetta omitted: their AF2Rank native rows contain -1 sentinel scores.

pilot_ptm_vs_tm_score.png

Each circle is a TM-stratified Rosetta decoy; stars are native structures. Colors denote the nine length/count-stratified targets.



```
$ plot_pilot.py
```

pilot_spearman_comparison.png

Mean target-wise Spearman correlation with reference TM-score; bars are 95% target-bootstrap intervals. Helico pTM is the primary score; the Helico composite additionally uses candidate/output TM-score.

